

Computer Networking



谢逸
中山大学·计算机学院
2024. Fall



学术经历

教学科研: 中山大学计算机学院

访问学者: Deakin U, Melbourne, Australia.

Georg Mason U, Fairfax, USA.

本/硕/博: 中山大学

联系方式

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● 教学:

- 本科: 计算机网络、通信原理;
- 硕士/博士: 高级计算机网络、高级网络与信息安全

● 科研:

- 方向: 面向计算机网络的基础性研究。
- 目标: 探索通信网络的工作机理及潜在规律, 为下一代网络设计、管理及提升通信服务质量提供理论支持。
- 项目: 国家自然科学基金、华为合作项目
- 成果: 网络领域顶刊



研究工作



计算机网络体系结构、
通信协议、网络安全、
资源调控、质量保障、
用户体验、.....

算力调度、工业控制、
无人系统、移动计算、
远程教育、智慧城市、
.....

欢迎参与网络研究!
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《计算机网络》课程概要

● Prerequisite Courses

- Operating System
- Probability and Statistics
- Programming

● Successive Courses

- Principle of communication
- Network Security
- Wireless communication
- ...

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《计算机网络》课程概要			
章节	主要教学内容	学时	重点、难点
第1章 计算机网络和因特网	课程简介, 网络边缘, 网络核心, 分组交换中的时延、丢包和吞吐量, 面对攻击的网络, 计算机网络和因特网的历史	6	重点: 协议的概念, 参考模型, 协议层次, 各层设计问题 难点: 网络相关概念的理解, 如协议、服务、交换等, 计算机网络的体系结构及各层交互的封装过程
第2章 应用层	应用层协议的原理, 应用层协议的实现过程, 进程通信, HTTP协议的持续连接, Web应用和HTTP协议, FTP协议, 电子邮件协议, DNS协议, P2P文件共享, TCP和UDP套接字编程基础, C/S架构应用实例	8	重点: 应用层协议的原理, Web应用和HTTP协议, TCP和UDP套接字编程 难点: HTTP协议的持续连接, 应用层协议的实现过程, DNS协议, P2P协议
第3章 传输层	传输层的基本概念和服务, 多路复用和多路分解, UDP的实现方法和相关传输原理, 可靠数据传输原理, TCP连接, TCP拥塞控制	10	重点: 多路复用和多路分解、TCP协议 难点: TCP的拥塞控制
第4章 网络层: 数据平面	网络层工作原理, 网络层的转发, IP协议, 网络服务和模型, 通用转发和SDN	6	重点: 网络层工作原理 难点: IP协议与数据包的网络层转发
第5章 网络层: 控制平面	路由选择算法, OSPF协议, BGP协议, SDN的控制平面, ICMP协议, 网络管理和SNMP协议	8	重点: 路由选择算法 难点: 路由协议
第6章 链路层和局域网	链路层服务, 链路层协议, 差错检测方法, CRC校验, 滑动窗口协议, 多路访问协议, 链路层原理, ARP协议, 以太网技术	10	重点: 多路访问协议, CRC校验, 以太网的工作原理 难点: 多路访问协议, 以太网的工作原理
第7章 无线网络	无线链路和网络特征, WiFi:802.11, 无线局域网	4	重点: WiFi:802.11协议, 无线局域网 难点: WiFi:802.11协议
考试	通过考试、进一步强化学生对计算机网络基础知识的理解和掌握	2	无

《计算机网络》课程概要

课程环节安排:

- 授课: 采用多媒体辅助课堂教学。
- 实验: 课堂教学与另外单独设置的计算机网络实验课程相配合。
- 作业: 每章均有作业。
- 考试: 期末闭卷考试, 内容涵盖课堂上所讲授的内容, 包括基本概念、计算和分析等题目。

《计算机网络》课程概要	
教学方法:	
● 课堂授课: 以课堂讲解基本概念和原理为主, 作业为辅。	
● 多媒体教学: 利用PPT呈现教学内容, 提高教学效率。	
● 课堂讨论: 对教学内容的重点, 通过课堂讨论加深理解。	
● 作业讲解: 讲解作业问题, 总结作业情况。	
● 答疑: 每周2小时办公时间答疑。	
■ 作业: 直接联系助教;	
■ 理论: 每周三中午(12点-14点, 学院楼A425), 请提前约定;	

《计算机网络》课程概要

教材与参考书

- 教材名称: 计算机网络: 自顶向下方法(原书第8版)
 - 作者: [美] James F.Kurose, Keith W.Ross 著, 陈鸣 译
 - 出版社: 机械工业出版社
- 主要参考书目
 - 《计算机网络》第8版, 谢希仁编著, 电子工业出版社, 2021.
 - 《Computer Networks》5th edition(英文版), A. S. Tanenbaum, 清华大学出版社, 2011.

《计算机网络》课程概要

成绩评定方式

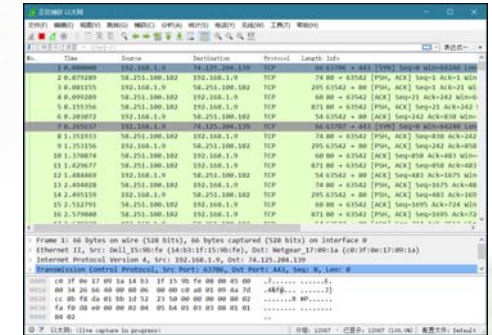
- 考核方式：考试
- 成绩录入方式：百分制
- 期末考试：闭卷，占总成绩比例60%。
- 平时成绩：书面作业，约5-7次，占总成绩比例35%；随堂考勤约5次，占总成绩比例5%。

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常用工具

Experimental tool: Wireshark

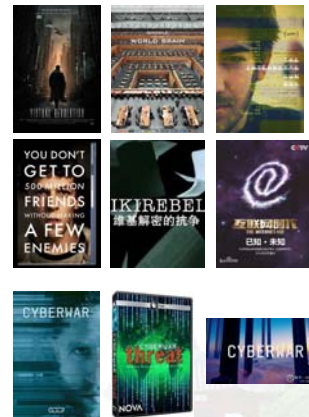
- Wireshark is a free and open source packet analyzer. It is used for network troubleshooting, analysis, software and communications protocol development, and education.
- <https://www.wireshark.org/>



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课外读物

- Recommended video:
 - BBC: "Virtual Revolution"
 - BBC: "Google and the World Brain"
 - "The Internet's own boy"
 - "The Social Network"
 - "维基解密的抗争"
 - CCTV: "互联网时代"
- Cyberwar season 1
- Cyberwar threat
- ...



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Miscellaneous

- Schedule: 18 Weeks vs. 500 PPTs ~ 50 pages/week
- Teaching pattern:
 - in class: We will only focus on **key points**, and encourage students to **question and discuss** some interesting topics.
 - after class: Students have to **read, think and prepare** the issues for class discussion. Otherwise, it will be difficult for passing the tests.
 - My advice:
 - Please show your questions in the class;
 - Please focus on "Why", "How";
 - please do NOT rely on your memory!

What can you get from this course?

- Basics of Data Communications;
- Ability to observe, think and solve problems. (More important)

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Some things about the experimental class

- Experimental classes start from the first week;
- Please complete the grouping for the experimental class **before this Wed.: 4 students per group**
 - An **online table** will be created for it and distributed via **Enterprise WeChat**.

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Contact Information

- Teaching Assistant: 汪昊元
- 课程微信群(企业微信):



●理论课PPT讲义:

https://gaia.cs.umass.edu/kurose_ross/index.php



<https://pan.sysu.edu.cn/link/AABA104C02E83E473A9BFB2E7E3DE8F96A>
 文件夹名: 理论课课件
 有效期限: 2025-01-20 21:30
 提取码: sLsT

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第1节 内容:

- 开设计算机网络课程的原因
- 现代数据网络的发展历史

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为什么需要学习计算机网络?

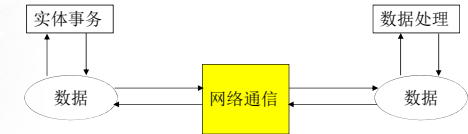
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- ◆ 人类历史上影响最深远的发明创造之一；
- ◆ 互联网促使现实物理世界和虚拟数字世界走向深度融合。



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- ◆ 物理世界中的事务处理方式转向以数据为基础、通信为支撑的模式。
- ◆ 网络空间全球一体化成为不可逆转的发展方向。



网络是全球不可或缺的关键基础设施之一

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- 网络是连接世界的重要手段
- 网络是信息传递的唯一途径
- 通信深植于现代日常生活中
- 涉及个人隐私、社会稳定、国家安全



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- 网络不可用时会怎么样？

■ 当网络不可用时：

数据无法传递	消息无法发送	网络资源失联	电商不可访问	游戏无法继续
视频无法观看	图像识别失效	NLP停止工作	机器人罢工	人工智能停顿
无人机停飞	直播无法上网	金融交易中断	远程手术中断	远程会议中断
工控国防瘫痪	影响人类生活	影响社会稳定	威胁国家安全	

在所有IT相关的技术应用中，没有一种具有计算机网络一样的战略地位！

数据通信网络面临的挑战

- 网络业务与应用快速更新迭代；
 - 多媒体、VR、游戏；
 - 人工智能、分布式算力；
 - 教育、医疗、金融、国防；
 - 工业、无人系统、物联网；
 -
- 网络体系结构不适应现代需求；
 - 通信质量需求；
 - 用户体验需求；
 - 数据业务需求；
 - 通信安全需求；
 -

网络技术需要与时俱进



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现代网络的发展简史

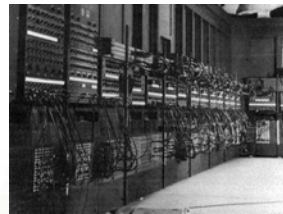
- Part I Connecting Hosts
- Part II Connecting Networks
- Part III Connecting the World
- Part IV Connecting everything

了解过去、认识现在、创造未来

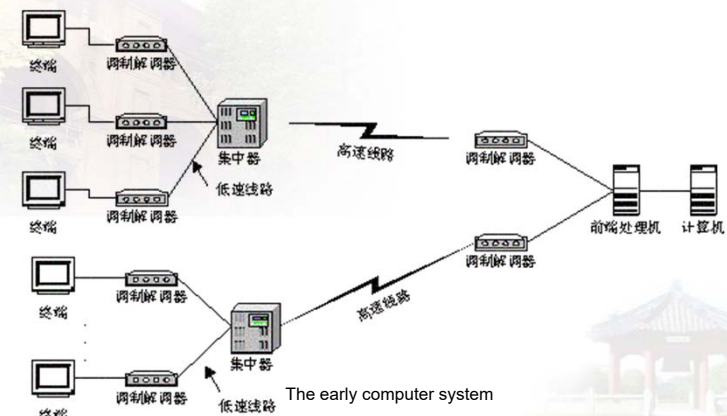
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Connecting Hosts (1940-1972)

- 1940, George Stibitz used a Teletype machine to send instructions or a problem set from his Model at **Dartmouth College** to his Complex Number Calculator in **New York** and received results back by the same means.
- 1946, The first electronic computer, ENIAC, was born in the United States.
- For the next 20 years people have been studying **the fusion of computers and communication systems**.

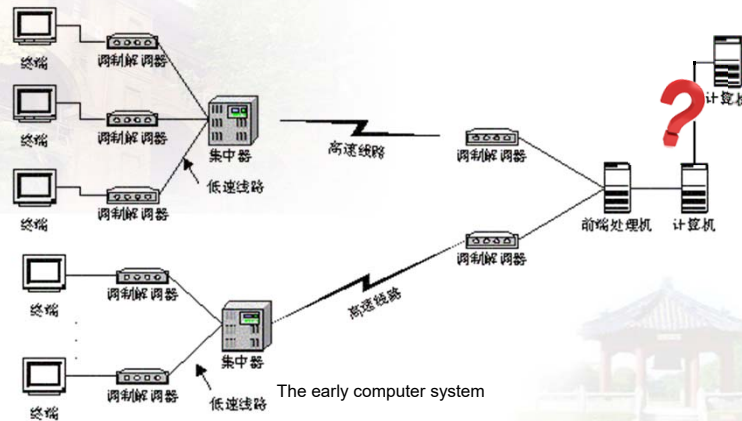


Connecting Hosts (1940-1972)



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Connecting Hosts (1940-1972)



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Connecting Hosts (1940-1972)

The story began in the 1950s. It is a star-studded age in which there is a group of Internet pioneers admired by future generations.



INTERNET HALL of FAME PIONEERS



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Connecting Hosts (1940-1972)

- 1957, Sputnik launched
- 1958, Advanced Research Projects Agency (ARPA) in the Department of Defense (DoD) was formed as response

Research on packet networks during 1959-1966:

- Lenoard Kleinrock: *Information Flow in Large Communication Networks*, PhD Thesis, 1961.
- Paul Baran: "On Distributed Communications Networks", 1961.
- Larry Roberts/Tom Marill publish first paper on network experiments: "Toward a Cooperative Network of Time-shared Computer" 1966.

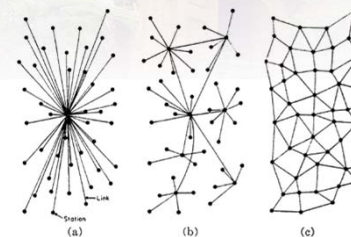


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Connecting Hosts (1940-1972)

The motivation of connecting terminals:

- ... consider the synthesis of a communication network which will allow several hundred major communications stations to talk with one another after an enemy attack...



... we must use a switching scheme able to find any possible path that might exist after heavy damage ... we shall explore the possibilities of building a "real-time" data transmission system using store-and-forward techniques...

Paul Baran, "On distributed communications networks," 1962

This directly derived the packet switching and store-and-forward.

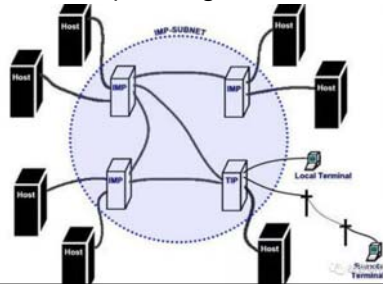
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• Connecting Hosts (1940-1972)

■ The motivation of connecting terminals:

- II. In 1966, Robert Taylor decided that ARPA should **link the existing computers at ARPA-funded research institutions together**. This would allow everybody on the network to **share** computing resources and results. Larry Roberts ("father of the ARPANET"), managed the project.

This directly derived the ARPA net.



• Connecting Hosts (1940-1972)

Sputnik launched → ARPA → packet switching theory
 share computing resources → Interconnected Network

Internet in 3 three years

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• Connecting Hosts (1940-1972)

- 1959-62, a **mathematical theory of packet networks** is created at MIT by **Leonard Kleinrock**



Professor Leonard Kleinrock is Distinguished Professor of Computer Science at UCLA. Known as a "**Father of the Internet**," he developed **the mathematical theory of packet networks**, the technology underpinning the Internet, while a graduate student at MIT. This was in the period 1960–1962, nearly **a decade before the birth of the Internet** which occurred in his laboratory when his Host computer at UCLA became the first node of the Internet in September 1969. He wrote **the first paper** and published **the first book** on the subject; he also directed the transmission of **the first message ever to pass over the Internet**.



<https://www.computer.org/profiles/leonard-kleinrock>

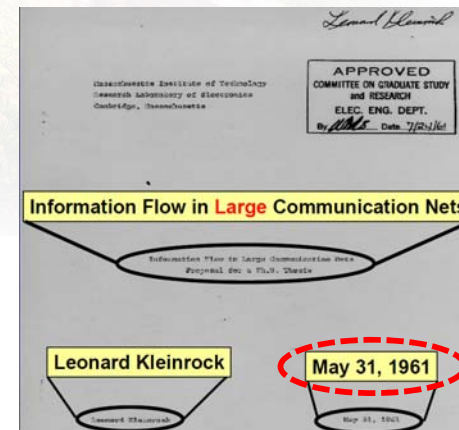
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• Connecting Hosts (1940-1972)

Leonard Kleinrock's
Ph.D thesis:

Information flow in large communication nets

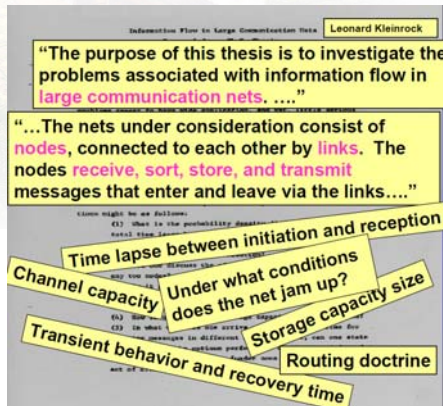
Packet switching theory is proposed



The idea, concept, implementation and technical details of **packet switching** were first proposed.

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• Connecting Hosts (1940-1972)

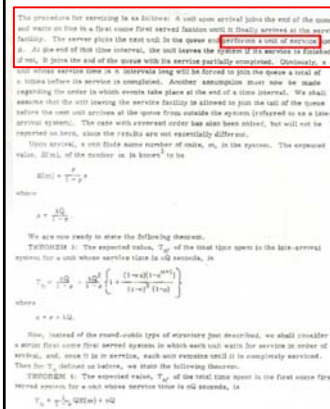


Keywords (1961):

- Information flow,
- Large communication nets,
- nodes,
- links,
- receive,
- transmit,
- store,
- sort,
- storage capacity,
- Channel capacity,
- jam up,
- recovery,
- routing,
- transient behavior,
- ...

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• Connecting Hosts (1940-1972)



The procedure for servicing is as follows: A unit upon arrival joins the end of the queue, and waits on line in a first come first served fashion until finally arrives at the service facility. The server picks the next unit in the queue and performs a unit of service upon it. At the end of this time interval, the unit leaves the system if its service is finished; if not, it joins the end of the queue with its service partially completed.

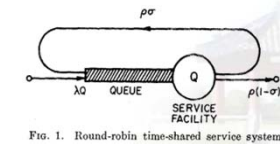


FIG. 1. Round-robin time-shared service system

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• Connecting Hosts (1940-1972)

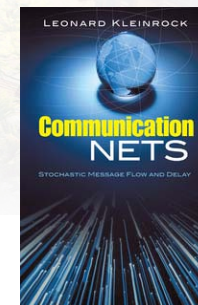
Leonard Kleinrock

Created a mathematical theory of data networks

- Channel capacity limited
- Mean response time as key metric
- Optimal assignment of channel capacity
- Choice of priority queueing discipline
- Concept of breaking messages into fixed size blocks
- Choice of routing procedure
- Design of topological structure
- Developed underlying principles of data networks that are the basis of the Internet

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• Connecting Hosts (1940-1972)



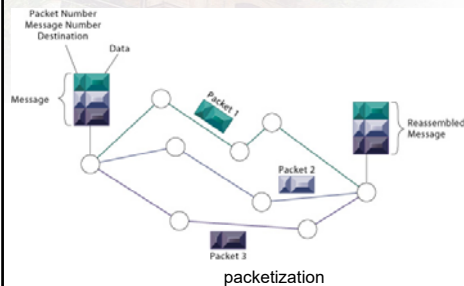
Communication nets: stochastic message flow and delay
By: **Leonard Kleinrock**
1964

Contents	
1. The Independent Assumption	7.1 Specific Network Descriptions
1.1 The Effect of the Independent Assumption	7.2 Simulation Results
1.2 Summary	7.3 Additional Theorems on Alternate Routing Procedures
2. Some new results for multiple-channel systems	7.4 Conclusions
2.1 A General Representation for the Uniform Traffic	
2.2 Optimum Routing of Traffic in a Single-Node Facility	
2.3 Traffic Relations among Rate, Capacity, and Storage Delay	
2.4 Optimum Assignment of Channel Capacity	
2.5 Conclusions and Extensions	
3. Waiting times for certain queue disciplines	
3.1 Priority Queueing	
3.2 A Queue with Loss	
3.3 Space-Shared Service	
3.4 Conclusions and Extensions	
4. Random routing procedures	
4.1 Uniform Random Routing	
4.2 The Average Path Length	
4.3 Examples of Flow Controlled by the Nodes	
4.4 The Fundamental Communication Theorem	
4.5 Conclusions	
5. Simulation of communication nets	
5.1 The Simulation Problem	
5.2 The Simulation Method	
5.3 The Simulation Results	
5.4 The Simulation Conclusions	
6. Appendixes	
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Connecting Hosts (1940-1972)

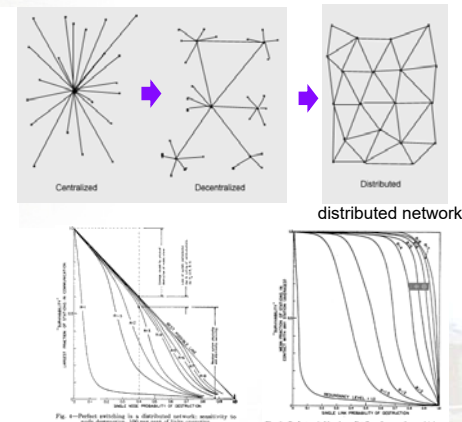
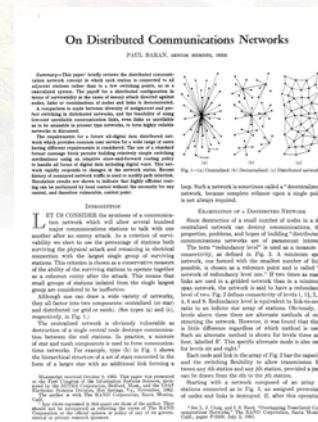
- 1962, Pau Baran suggests transmission of data using fixed size **message blocks**
- 1964, Pau Baran publishes reports "on **distributed communications**"



he worked on developing a communications system that could survive the damage of a nuclear weapon. Originally, he called the process "**message blocks**." Other scientists including Donald Davies, who would later change the name to "**packet switching**," had also come to a similar conclusion at the same time.

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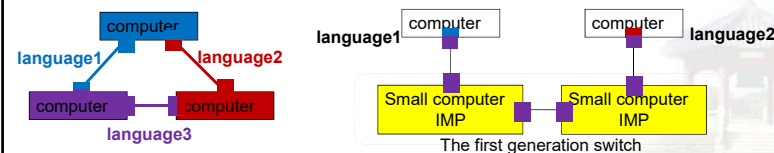
Connecting Hosts (1940-1972)



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Connecting Hosts (1940-1972)

- Wes Clark suggested that Roberts employ **small computers at each site to handle networking functions** and leave the host computers alone.
- All of the small computers could speak the same language** which would facilitate communication between them.
- Each host computer would only have to adapt its language once in communicating with its small computer counterpart.
- Each host computer would be connected to the network via its small computer which would act as a sort of gateway.
- The small computers could also remain under more direct ARPA control than were the large host computers.



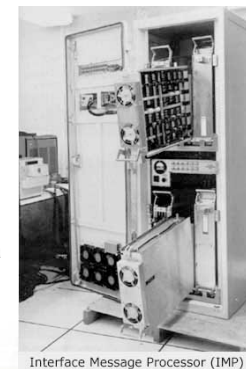
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Connecting Hosts (1940-1972)

- The small computer is called **Interface Message Processors (IMPs)** in 1967, which became the **router/switch** later.

The **Interface Message Processor (IMP)** was the packet switching node used to interconnect participant networks to the ARPANET from the late 1960s to 1989. It was the first generation of gateways, which are known today as **routers**.

Data from: Wikipedia



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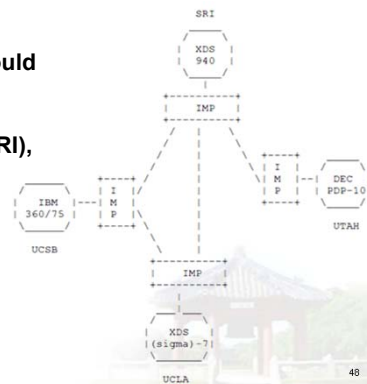
• Connecting Hosts (1940-1972)

- 1968, Roberts publishes ARPANET plan

- Roberts decided that the network should start out with **four sites**:

- **UCLA**,
- **the Stanford Research Institute (SRI)**,
- **the University of Utah**,
- **and UC Santa Barbara**.

- This would be **the core of the later Internet**.



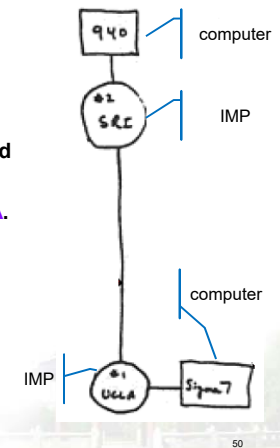
• Connecting Hosts (1940-1972)

- The design of the first network:

- Host computer would be attached to a specialized computer called **Interface Message Processor, IMP**.
- An IMP would be connected to one another.
- An IMP had to be able to communicate with Attached host and Other IMP

- In August 1969, they delivered **the first IMP to UCLA**. A month later, **the second was delivered to SRI**.

- **The two were connected and the ARPANET was born.**

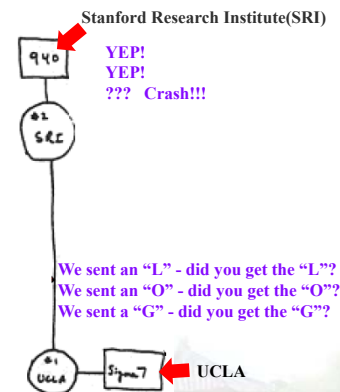


• Connecting Hosts (1940-1972)

- On **Oct. 29, 1969**, Leonard Kleinrock and his graduate student Charley Kline wanted to send a transmission **from UCLA's computer to another computer at Stanford Research Institute through ARPANET**.
- The message sent by Kleinrock and Kline was intended to be **"login."** However, the system crashed as soon as they typed **the second letter "o"**. It took an hour to send the whole word.
- **"lo"** cemented its place in the Internet's history.
- **"L" and 'O' is 'hello,' and a more succinct, more powerful, more prophetic message we couldn't have wished for"**—Kleinrock

The Internet is BORN at UCLA on October 29, 1969 !

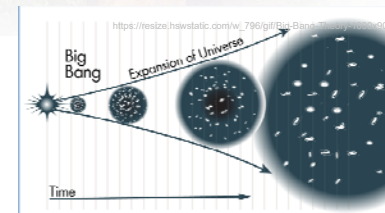
Two years later, in 1971, the **first email** was sent by MIT researcher Ray Tomlinson — which was also the first time the **"@"** sign was used to designate a specific recipient of a message.



• Connecting Hosts (1940-1972)

- ***UCLA** becomes a **harbinger** of the modern era as it launches ARPANET, the **precursor** to the modern internet, from its Measurement Center in September, 1969.

- UCLA** becomes the **first "node"** of the internet, and the first message **"LO"** of **"LOGIN"** goes from UCLA to the Stanford Research Institute, Internet Node #2, on October 29th, 1969.



In that moment, the Internet was born, and with it, an almost **limitless number of possibilities for the future.**



*<https://100.ucla.edu/timeline/the-internets-first-message-sent-from-ucla>

• Connecting Hosts (1940-1972)

- 1969 Steve Crocker **RFC #1** Host-Host Protocol and the NWG

the 1st RFC: Steve Crocker, "Host Software," known as the **Network Control Protocol (NCP)**, 7th April, 1969.

RFC Index

0001 Host Software S. Crocker [April 1969]



50+ years :

9654 Online Certificate Status Protocol (OCSP) Nonce Extension H. Sharma [August 2024]

<https://www.rfc-editor.org/rfc-index.html>



• Connecting Hosts (1940-1972)

- 1970 ARPANET spans US: UCLA <-> BBN
- 1972 Ray Tomlinson introduce net email and "@"
- 1972 First public demo of ARPANET



- Network Working Group (NWG) adopted a **layered** approach to the specification of communications protocols.
- A Bhushan, RFC 114, A **F**ile **T**ransfer **P**rotocol (April 16, 1971)
- T O'Sullivan, NWG RFC 158, **TELNET** Protocol (May 19, 1971)

现代网络的发展简史

- Part I Connecting Hosts
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• Connecting Networks (1972-1989)

- 1972 **Additional packet-switching networks** besides ARPAnet came into being;
 - ALOHAnet, a satellite network linking together universities on the Hawaiian islands;
 - Telenet, a BBN commercial packet-switching network based on ARPAnet technology;
 - Tymnet; and Transpac, a French packet-switching network.

The number of networks was beginning to grow.

Once again, with perfect hindsight one might now see that the time was ripe for developing an encompassing architecture for **connecting networks** together.

- 1973 Cerf and Kahn design TCP

- [illegible]

- 1973 ARPA deploys SATNET – 1st international connection

- Figure 13: July 1975
-
- Legend:
- BIP
 - TIP
 - SATELLITE CIRCUIT

- **1973** Bob Metcalfe develops **Ethernet** idea

-
- Diagram illustrating a TAP interface for a token ring network. A TAP is connected to a TRANSFUSER (marked with an X). This is connected via an INTERFACE CABLE to a STATION (marked with an I). The STATION is connected to an INTERFACE CONTROLLER (marked with a C). The INTERFACE CONTROLLER is connected to a bus labeled THE ETHER. The bus has three other stations connected to it and ends with a TERMINATOR.

A 1976 drawing by R. Metcalfe of the Ethernet concept
(from Charles Spurgeon's Ethernet Web Site)

**Ethernet protocol
is designed**

• Connecting Networks (1972-1989)

- 1977 Cerf and Kahn demonstrate interconnection of networks **using IP** by interconnecting ARPANet, SatNet, Ethernet, and PRNET.
- 1977 BBN develops **first Internet Routers** in collaboration with Stanford
- 1978 **TCP splits into TCP and IP** driven by Danny Cohen, David Reed and John Schoch to support real-time traffic. This allows the creation of UDP.

TCPv1; TCPv2; TCP/IPv3 (splitting TCP into TCP and IP), and, in 1978, they settled on IP version4.

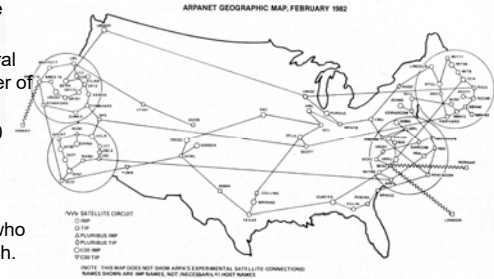
IP as originally designed had an **eight bit** networking field which would be sufficient for at maximum 256 networks



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1982: the ARPANET community grows

- As the ARPANET entered its **second decade**, it was still largely confined to the United States.
- Academic institutions depended on federal funding to join the network, so the number of nodes expanded slowly.
- By 1982, the network only had about 100 nodes. But that was enough to support a vibrant online community.
- Long before Facebook and Twitter, ARPANET allowed computer scientists who had access to the network to stay in touch.
- A new bulletin board system called **Usenet** was invented in 1980 and caught on quickly. Usenet was organized by topic, allowing users to swap programming tips, recipes, jokes, opinions about science fiction, and much more.



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1984: ARPANET becomes the internet

- Originally, the entire ARPANET was managed by the military.
- But network operators realized that a centralized network would eventually become unmanageable if it continued to grow.
- They decided that the network should be reorganized as a **decentralized "network of networks."**
- Under this scheme, different networks would be controlled by **different organizations**, but **all the networks able to communicate using shared standards, forming a shared "internet."**
- The military asked the computer scientists **Robert Kahn** and **Vint Cerf** to develop new networking standards to make this possible.
- The result was a set of standards known as **TCP/IP**.



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1984: ARPANET becomes the internet

- These standards specified **the basic format of data packets** transmitted across the internet.
- On **January 1, 1983**, the ARPANET switched to using TCP/IP, marking **the birth of the modern internet**.
- The switch to TCP/IP didn't make much difference from a user perspective — applications like email and Telnet worked about the same as they had before.
- But the new standard paved the way for much **faster network growth by lowering the barrier to entry for new networks**.
- One of the first new networks to connect to the new internet was **CSNET**, which was funded by the National Science Foundation to link computer science departments across the country.

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● Connecting Networks (1972-1989)

- 1985 Paul Mockapetris, Jon Postel and Craig Partridge **create the Domain Name system (DNS)**, the **first domain** is registered: symbolics.com
- 1985 White House, Science Council (WHSC) Committee on Research in Very High Performance Computing. The C++ was released.
- 1986 **Internet Engineering Task Force (IETF)** and Internet Research Task Force (IRTF) comes into existence
- 1987 Number of hosts breaks 10,000
- 1987 **The first email was sent successfully in China**
- 1989 Number of hosts breaks 100,000



It is also worth noting that in the 1980's (and even before), researchers were also developing a "competitor" technology to the Internet architecture. These efforts have contributed to the development of the **ATM** (Asynchronous Transfer Mode) architecture, a connection-oriented approach based on the use of fixed size packets, known as cells.

● Connecting Networks (1972-1989)

- 1990 ARPAnet is terminated on Feb. 28, 1990 and migrated to NSFNET
- 1991 **High-Performance Computing Act** that derives the later High Performance Computing and Communications (HPCC) Program
 - the purpose of this act is to help ensure the continued **leadership** by the United States and high-performance computing and its applications.
 - The goal of the HPCC Program is to accelerate the development of **future generations** of high performance computers and networks and the use of these resources in the Federal government and throughout the American economy. Scalable high performance computers, advanced high speed computer communications networks, and advanced software are critical components of a new **National Information Infrastructure (NII)**. This infrastructure is essential to our **national competitiveness** and will enable us to strengthen and improve the civil infrastructure, digital libraries, education and lifelong learning, energy management, the environment, health care, manufacturing processes and products, national security, and public access to government information.

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NSFNET: The first internet backbone

- During the 1980s, the National Science Network funded several supercomputing centers around the United States. And in 1986 the agency created a TCP/IP-based network called NSFNET to **link those supercomputing centers together** and allow researchers across the country to use them.
- The primary goal was to allow computer science researchers to log into the supercomputers and perform academic research.
- But NSF decided not to limit NSFNET to that purpose, allowing the network to be used for a wide variety of academic purposes.
- As a result, **the NSFNET became the internet's "backbone,"** the high-speed, long-distance network that allowed different parts of the internet to communicate.
- Schools that didn't have a direct connection to the NSFNET worked together to build **regional networks** that linked them to each other and to the nearest NSF node.
- This shows the NSFNET as it existed in 1992. By this time, there were 6,000 networks connected to NSFNET, with a third of them located overseas.

• Connecting the World (1990-2000)

- 1991 NSF opens Internet to commercial use
- 1991 First Webpage (Tim Berners-Lee)

On August 6, 1991, without fanfare, British computer scientist Tim Berners-Lee published the first-ever website while working at CERN, the huge particle physics lab in Switzerland.



This computer was used at CERN by British scientist Tim Berners-Lee to devise the World Wide Web (WWW).



Tim Berners-Lee, Inventor of the Web, poses in front of the first World Wide Web Server.

<https://www.history.com/news/the-worlds-first-web-site>

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• Connecting the World (1990-2000)

- The motivation of the development on the Web

Tim Berners-Lee later stated: “In those days, there was different information on different computers, but you had to log on to different computers to get at it. Also, sometimes you had to learn a different program on each computer.”

Berners-Lee didn't try to cash in on his invention and rejected CERN's call to patent his Web technology. He wanted the Web to be open and free so it could expand and evolve as rapidly as possible. As he later said, “Had the technology been proprietary, and in my total control, it would probably not have taken off. You can't propose that something be a universal space and at the same time keep control of it.”

<http://info.cern.ch/>

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• Connecting the World (1990-2000)

- 1993 Marc Andreessen Mosaic browser

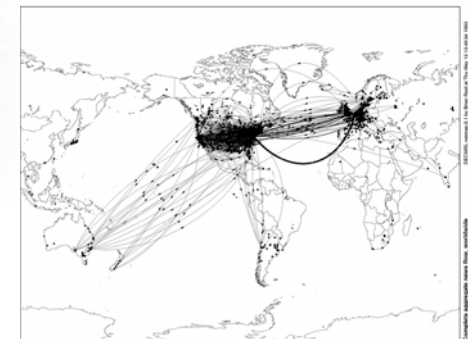
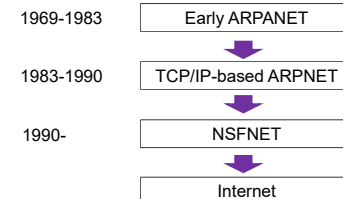
- NCSA Mosaic was neither the first web browser (first was the World Wide Web of Berners-Lee) nor the first graphical web browser (it was preceded by the lesser-known Erwise and ViolaWWW), but it was the web browser credited with popularizing the World Wide Web
- Mosaic was also a client for earlier protocols such as FTP, NNTP, and gopher.



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The internet becomes a global network

- In 1993, the internet was still dominated by the United States but it was becoming a truly global network.
- This is a map of information flow on Usenet, an bulletin board application that allowed users to swap recipes, jokes, programming tips, and more.



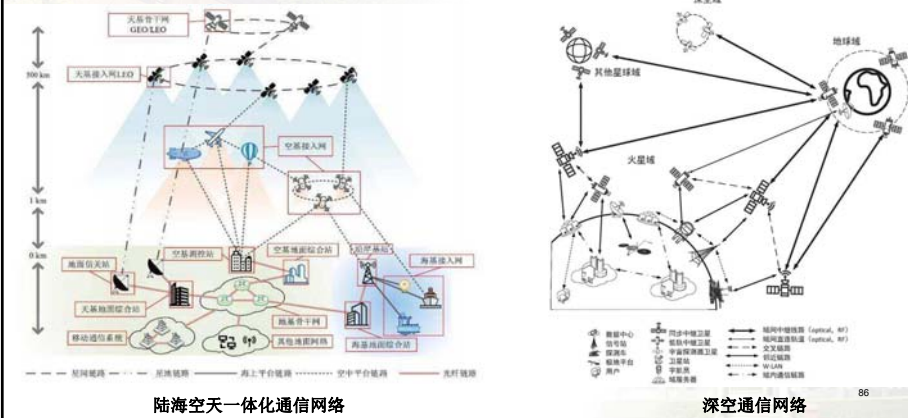
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Telephone, TCP/IP, Ethernet, Internet, WIFI, 6G, Mobile, Datacenter, Cloud, Fog, IoT ...



2030 6G 愿景



6G vision: Global coverage, all spectra, full applications, all senses, all digital, and strong security.

- 6G将构建人-物-物智慧互联、智能体高效互通的新型网络，在大幅提升网络能力的基础上，具备智慧内生、多维感知、数字孪生、安全内生等新功能。6G将实现物理世界人与人、人与物、物与物的高效智能互联，打造泛在精细、实时可信、有机整合的数字世界，实时精确地反映和预测物理世界的真实状态，助力人类走进人-物-物智慧互联、虚拟与现实深度融合的全新时代，最终实现“万物智联、数字孪生”的美好愿景。

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Networks are changing the world in evolution
A future with endless possibilities is coming ...

Are you ready?

Welcome to ComNet!

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Thanks

Q & A

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<https://cse.sysu.edu.cn/content/2462>